

AERO50005 TUTORIAL 3

Failure Questions

1. The micrograph in Figure 1 is of Mg-4Y-3Nd after tensile creep testing at 250°C. Examine the features and annotations to answer the following questions:

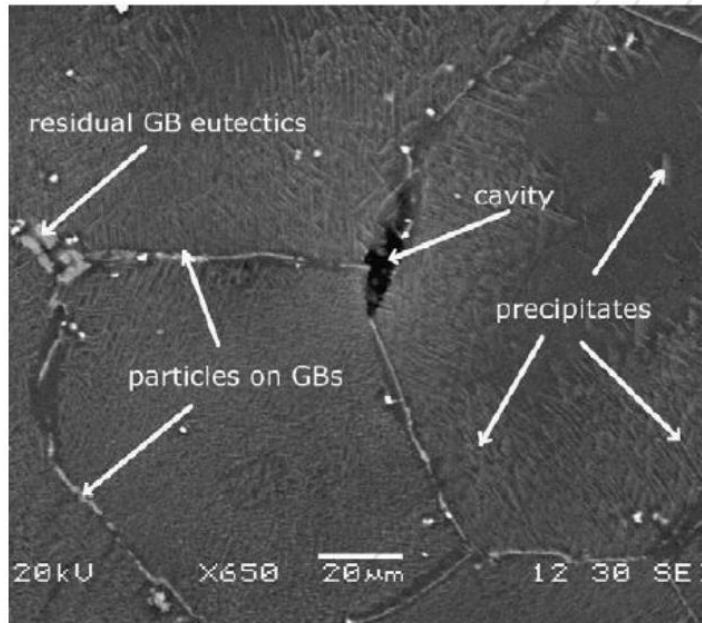


Figure 1. from Janik et al. *Mater. Sci. Eng. A* 528 (2011) 3105-3112.

- (a) Which microstructural features give this alloy creep resistance? Explain how each feature combats a specific creep mechanism.
- (b) If this alloy is heated to 350°C, it loses its creep resistance. Describe the changes in the microstructure that you would expect to be responsible for this.
- (c) By examining the creep damage in the micrograph, suggest the direction of the tensile axis during loading.
- (d) In a gearbox housing made from such an alloy, creep failure occurs when the total strain reaches 4%. Currently the housing operates at a stress of 50 MNm⁻² which generates a strain rate of 3.93*10⁻¹¹ s⁻¹ at 250°C. A new design requires the housing to operate at 70MNm⁻².
How long will it take the housing to fail at this stress level at 250°C?

Hint: “power law creep” equation for this material is:

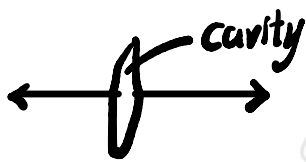
$$\dot{\epsilon} = A\sigma^5 e^{-\frac{Q}{RT}}$$

(a) $\frac{\beta'}{\beta}$ precipitates will become obstacles of the dislocation motions, hindered dislocation creep.

Meanwhile, particles on grain boundaries could prevent grain boundary sliding as microstructure want to maintain its continuity, give a resistance to the diffusion creep happened.

(b) When the temperature goes up, coarsening might happened to the precipitates in microstructure, therefore reduce the area of obstacles. Also, the precipitates on the grain boundary might ~~form other phases with Mg~~, diffusion creep resistance also gone.
also coarsening

(c) According to the cavity, creep direction should be



$$(d) Ae^{-\frac{Q}{RT}} = \frac{\dot{\epsilon}}{6^5} = 1.2576 \times 10^{-49}$$

$$\therefore \dot{\epsilon}' = 6^5 (Ae^{-\frac{Q}{RT}}) = 2.114 \times 10^{-40} \text{ (s}^{-1}\text{)}$$

$$\therefore \dot{\epsilon}' = \frac{\Delta \epsilon'}{\Delta t} \quad \therefore \Delta t = \frac{\Delta \epsilon'}{\dot{\epsilon}'} = 1.89 \times 10^8 \text{ s}$$

2. Use the Palmgren-Miner-Rule (Equation 1) to estimate whether or not a crack free 2014-T6 alloy (Fatigue data Figure. 2) will experience fatigue failure under the following conditions (assuming $\sigma_m = 0$): (1) $\sigma_a = 200$ MPa for 10^5 cycles and $\sigma_a = 100$ MPa for 10^7 cycles (2) $\sigma_a = 200$ MPa for 10^7 cycles then $\sigma_a = 100$ for 10^5 cycles and $\sigma_a = 300$ MPa for 10^4 cycles.

$$\sum_i \frac{n_i}{N_B} = 1 \quad (1)$$

$$(1) \sum_i \frac{n_i}{N_B} = \frac{10^5}{8 \times 10^5} + \frac{10^7}{9.3 \times 10^7} = 0.126 < 1 \quad \therefore \text{safe}$$

$$(2) \sum_i \frac{n_i}{N_B} = \frac{10^7}{8 \times 10^5} + \frac{10^5}{9.3 \times 10^7} + \frac{10^4}{3 \times 10^4} = 12.833 > 1$$

$\therefore$ fatigue failure

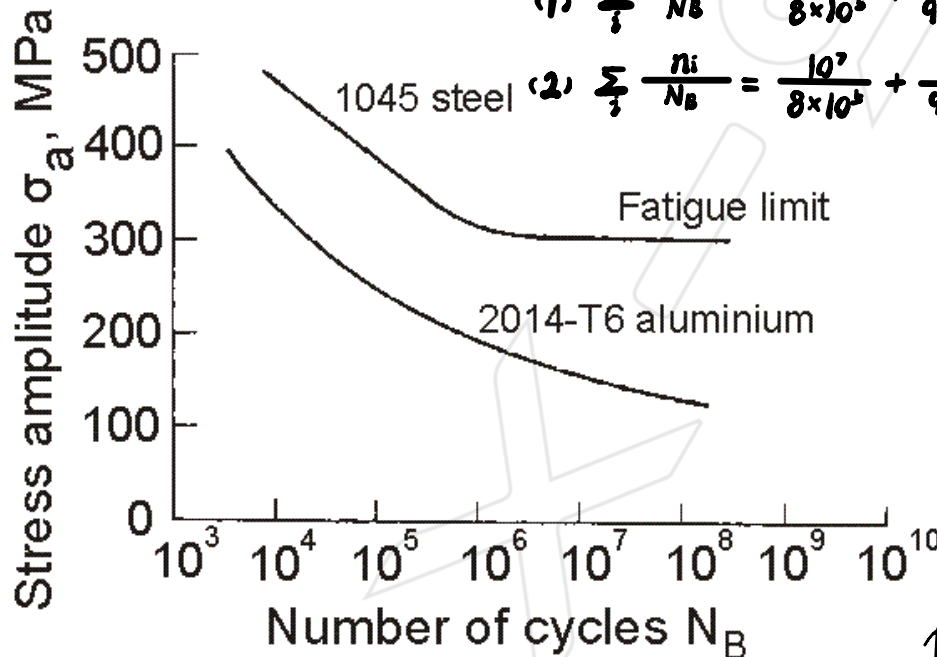
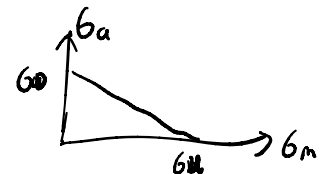
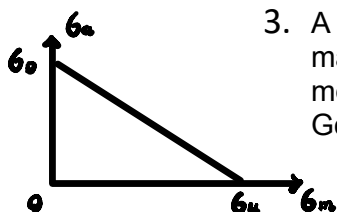


Figure 2. S-N curve



$$F_a = 220 \text{ kN}, F_m = -110 \text{ kN}$$

3. A 4340 steel bar is subjected to a fluctuating axial load that varies from a maximum of 330 kN tension to a minimum of 110 kN in compression. The mechanical properties of the steel are: $\sigma_u = 1010$ MPa; $\sigma_D = 510$ MPa. Using the Goodman law determine the bar diameter d to give infinite fatigue life.



$$\frac{\sigma_u}{\sigma_D} + \frac{\epsilon_u}{\epsilon_D} = 1 \quad \therefore \frac{1}{d} \left(\frac{F_a}{\sigma_D} + \frac{F_m}{\epsilon_D} \right) = 1 \quad \therefore d = 2 \sqrt{\frac{A}{\pi}} = 0.0262 \text{ m}$$

4. A mild steel plate is subjected to constant amplitude uniaxial fatigue loads to produce stresses varying from $\sigma_{\max} = 180$ MPa to $\sigma_{\min} = -40$ MPa. The static properties of the steel are $\sigma_0 = 500$ MPa, $\sigma_N = 600$ MPa, $E = 207$ GPa and $K_c = 100$ MPa $m^{1/2}$. If the plate contains an initial through thickness edge crack of 0.5 mm, how many fatigue cycles will be required to break the plate? Assuming an infinite wide plate, for which $Y = 1.12$ (Y is the geometry factor). For Ferritic-pearlitic steels, a general correlation gives: da/dN (m/cycle) = $6.9 \times 10^{-12} (\Delta K)^3$ (MPa $m^{1/2}$)³. Thus $A = 6.9 \times 10^{-12}$ (MPa $m^{1/2}$)³, $m = 3$, $a_0 = 0.005$ m. *all in MPa*

$$\Delta \sigma = 180 \text{ MPa}$$

$$\Delta K = Y \Delta \sigma \sqrt{\pi a}$$

$$K_c = Y \sigma_{\max} \sqrt{\pi a}$$

$$N_f = \int_{a_0}^{a_f} \frac{da}{A (\Delta K)^m} = \int_{a_0}^{a_f} \frac{1}{A Y^3 \Delta \sigma^3 \pi^{\frac{3}{2}}} a^{-\frac{3}{2}} da = \frac{-2}{A Y^3 \Delta \sigma^3 \pi^{\frac{3}{2}}} [a^{-\frac{1}{2}}]_{a_0}^{a_f} = 37051.1724 [a^{-\frac{1}{2}}]_{a_0}^{a_f} = 261000 \text{ (cycles)}$$

$$a_f = \frac{1}{\pi} \left(\frac{K_c}{Y\sigma_{\max}} \right)^2 = 0.0783193 \text{ (m)} = 78.319 \text{ (mm)}$$

$$N_f = \int_0^{N_f} dN = \int_{a_0}^{a_f} \frac{da}{A(\Delta K)^m}$$

Hint: use Paris law equation

5. (a) Describe the three types of oxidation kinetics that are common for metals. Explain, with the aid of sketches, the mechanisms that determine how the rate of oxidation depends on time in each case.
- (b) Laboratory tests on the oxidation of a Ni-superalloy aeroengine turbine blade at 1000°C gave an average oxide layer thickness of 500µm after 500 hours. Assuming that oxide layer growth is controlled by diffusion through the oxide layer, predict the oxide layer thickness after 300 hours at 1000°C.
- (c) Explain why alloying with aluminium can improve the oxidation performance of a Ni-superalloy.
- (d) Summarise how Ni-superalloys operate as turbine blades when the hot gas temperature exceeds the melting point of the alloy.

(a) parabolic growth. rate of oxidation is dependent of time;

linear growth. rate of oxidation is independent of time;

linear loss. rate of oxidation is independent of time.

(b) for Ni. corrosion is parabolic growth

$$\therefore \Delta m^2 = k_p \Delta t \quad \therefore \Delta x^2 \propto k_p \Delta t \quad \therefore \left(\frac{\Delta x_2}{\Delta x_1} \right)^2 \propto \left(\frac{\Delta t_2}{\Delta t_1} \right) = 0.6$$

$$\therefore \Delta x_2 = 0.774 \Delta x_1 = 387.30 \text{ (}\mu\text{m)}$$

(c) Because when coating with Al, some Al will diffuse into the Ni to form AlNi and the rest will form Al₂O₃ layer on the Ni-superalloy surface. improved its oxidation performance as oxides growth very slowly in Al

$$a_f = \frac{1}{\pi} \left(\frac{K_c}{Y\sigma_{\max}} \right)^2$$

$$N_f = \int_0^{N_f} dN = \int_{a_0}^{a_f} \frac{da}{A(\Delta K)^m}$$

Hint: use Paris law equation

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d. Ni₃Al phase has excellent high temperature strength

Single crystals gives excellent creep resistance

Al and Cr improve high T oxidation

Blade cooling

Thermal barrier coating